BUOY DATA

This page contains the real time readings from our four continuous monitoring water quality monitoring buoys in Lake Maurepas: the Amite (blue), Blind (black), Maurepas (purple), and Tickfaw (pink) buoys, as well as from our seven continuous monitoring chemistry buoys (yellow). By providing real-time data, our aim is to enhance the community's understanding of Lake Maurepas' current status and to elevate their awareness regarding water quality, water chemistry, and atmospheric conditions in the area. Click on the buoys in the interactive map below to learn more details about each one.

Enable map scroll

+

–

CONANP | Esri | TomTom | Garmin | SafeGraph | CGIAR | USGS | FAO | METI/NASA | EPA | NPS | USFWS   Powered by Esri

Data Disclaimer: Buoy data and sensor data displayed on this website are collected in real time from autonomous environmental monitoring buoys and are subject to sensor limitations, equipment malfunctions, communication errors, biofouling, and environmental interference. Routine maintenance and sensor checks are performed to support data reliability, and identified issues are addressed as soon as possible. However, live data may contain erroneous, incomplete, or anomalous values and should be interpreted with consideration of these limitations.

If viewing on a mobile device, scroll down on each dashboard to view the full list of parameters.

Water Quality Buoys

The water quality parameters monitored by these buoys include CO2, temperature, turbidity, specific conductivity, salinity, dissolved oxygen concentration (mg/L and % saturation), and pH. The Blind Buoy is also fit with a weather station that monitors atmospheric conditions in addition to water quality. This includes air temperature, rain total, rain intensity, dew point, barometric pressure, relative humidity, wind speed, wind gust speed, and wind direction. The Amite, Tickfaw, and Maurepas buoys are also fit with a wave sensor, which reports wave height. For more information regarding the parameters monitored by these buoys, see the table below. All parameters are measured every 15 minutes, starting on the date of deployment (wave sensors: 4/8/2025, all other sensors: 1/31/2024).

CO_2

Units: Parts per million by volume (ppmV)

Buoys: All

Description: Partial pressure of CO2 gas dissolved in the surface water.

Temperature

Units: Fahrenheit (F)

Buoys: All

Description: Temperature of the water.

Turbidity

Units: Nephelometric Turbidity Unit (NTU)

Buoys: All

Description: The opaqueness of the water due to the presence of suspended solids.

Specific Conductivity

Units: Microsiemens per centimeter (mS/cm)

Buoys: All

Description: The electrical conductance of 1 cubic centimeter of the water at 25 degrees Celsius.

Salinity

Units: Parts per thousand (ppt)

Buoys: All

Description: The number of grams of salt per 1000 grams of water.

DO Concentration

Units: Milligrams per liter (mg/L)

Buoys: All

Description: The concentration of free oxygen (as a gas) dissolved in the water.

DO Concentration

Units: Percent Saturation (%)

Buoys: All

Description: Oxygen saturation as the percentage of dissolved O2 concentration relative to that when completely saturated at the temperature of the measurement depth.

pH

Units: None

Buoys: All

Description: Measurement of how acidic or alkaline the water is. A pH of 7 is neutral, a pH of <7 is acidic, and a pH of >7 is basic.

Air Temperature

Units: Fahrenheit (F)

Buoys: Blind

Description: Temperature of the air.

Rain Total

Units: Millimeters (mm)

Buoys: Blind

Description: Accumulated rain reading. Resets to 0 millimeters at 17:00.

Rain Intensity

Units: Millimeters per hour (mm/hr)

Buoys: Blind

Description: Average precipitation rate over the last 1 hour.

Dew Point

Units: Fahrenheit (F)

Buoys: Blind

Description: A measure of atmospheric moisture. It is the temperature to which air must be cooled in order to reach saturation (assuming air pressure and moisture content are constant). A higher dew point indicates more moisture present in the air.

Barometric Pressure

Units: Millibar (mbar)

Buoys: Blind

Description: The pressure of the atmosphere as indicated by a barometer (air pressure).

Relative Humidity

Units: Percent (%)

Buoys: Blind

Description: A ratio of the amount of atmospheric moisture present relative to the amount that would be present if the air were saturated. Since the latter amount is dependent on temperature, relative humidity is a function of both moisture content and temperature.

Wind Speed

Units: Miles per hour (mph)

Buoys: Blind

Description: The average speed of air flow past a horizontal point. The averaged output reading is based on the previous 15 minutes of wind data.

Wind Gust Speed

Units: Miles per hour (mph)

Buoys: Blind

Description: The magnitude of the maximum gust measured over a 15-minute averaging data output period. Gust is determined from a rolling 3 second average throughout the 15-minute period, and is reset at the end of those 15 minutes.

Wave Height

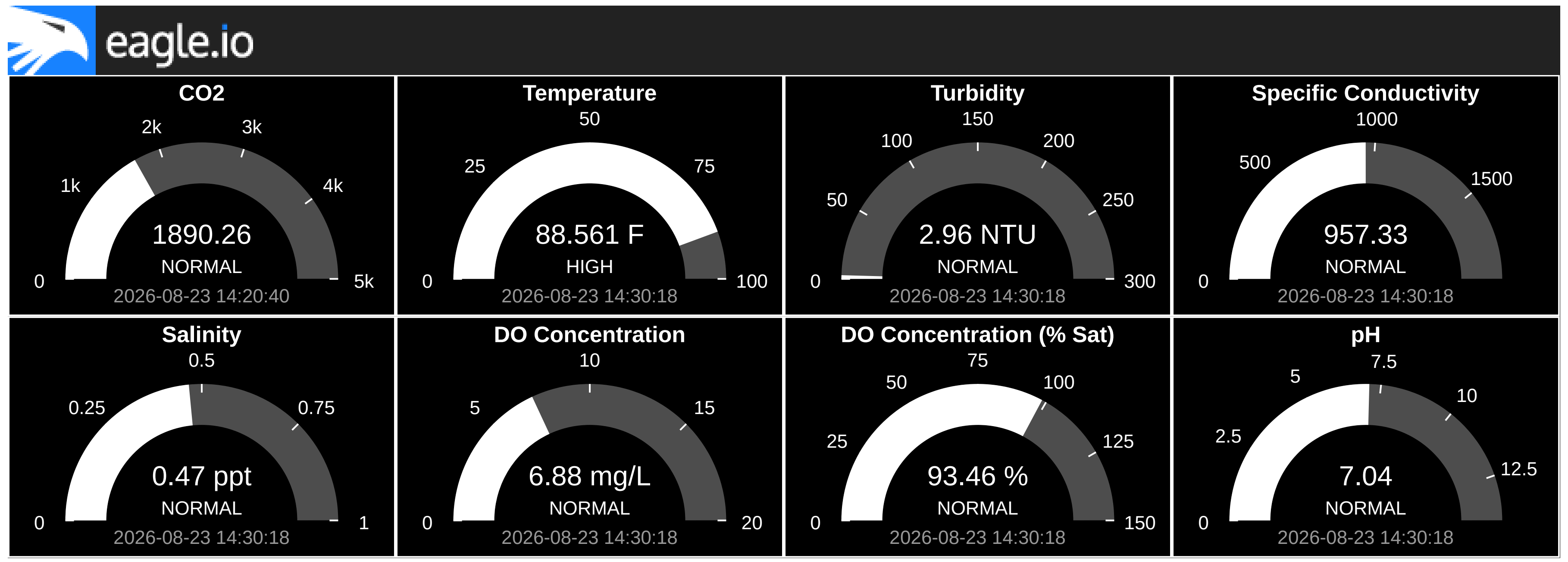
Units: Wave Height (m)

Buoys: Maurepas, Tickfaw, Amite

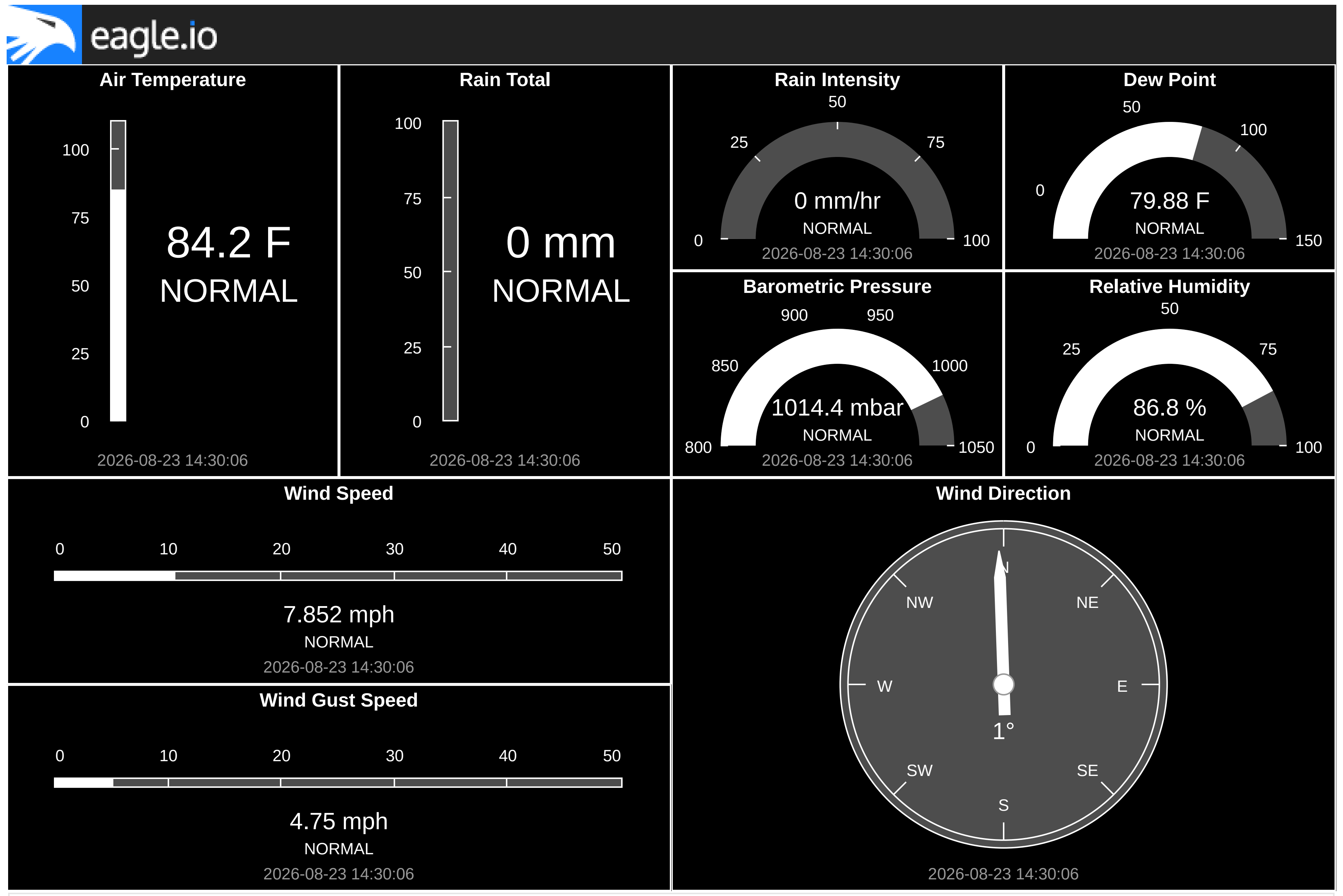
Description: Height of wave from trough to crest.

Amite Buoy Water Quality Data

Blind Buoy Water Quality Data

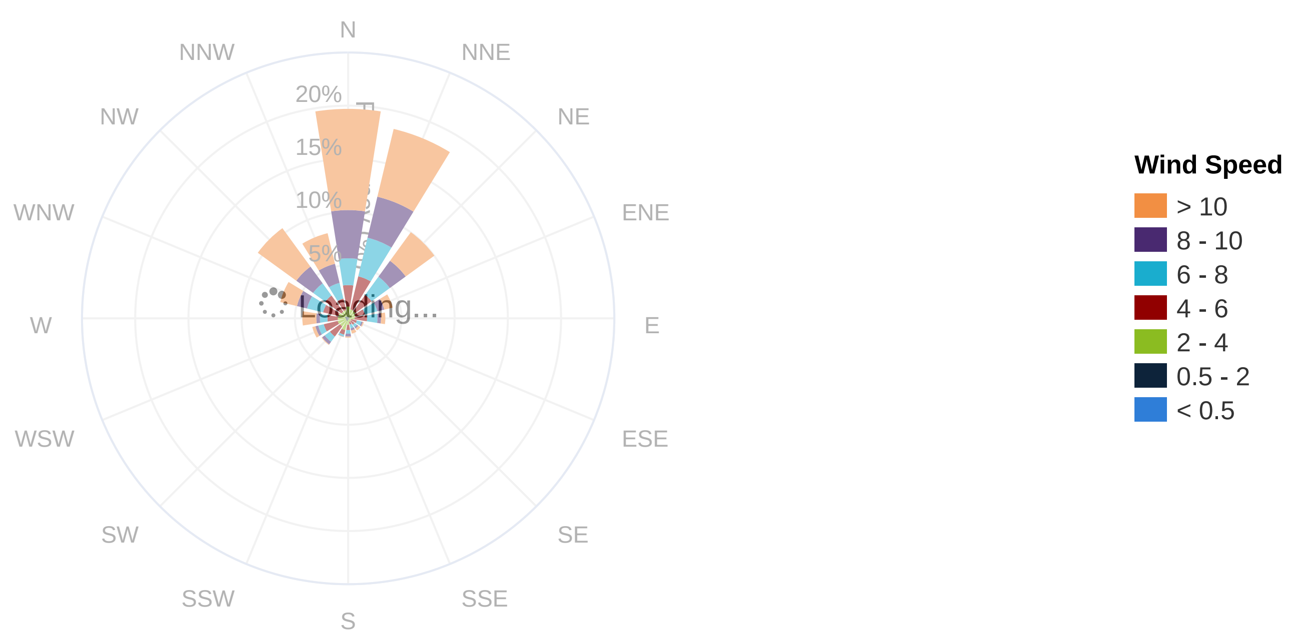


Blind Buoy Weather Station Data

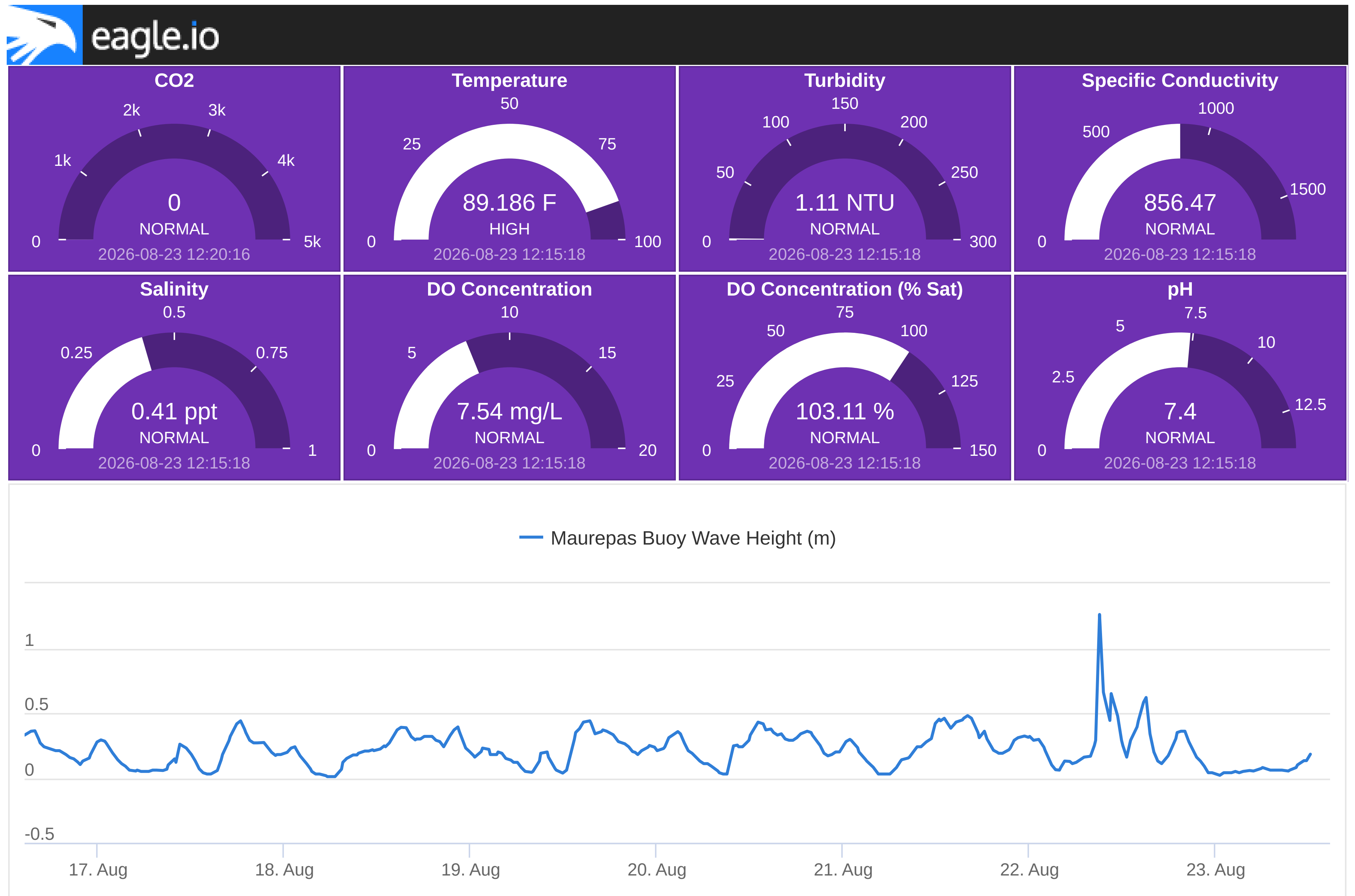


Wind Rose Chart: Historic Wind Speed and Direction

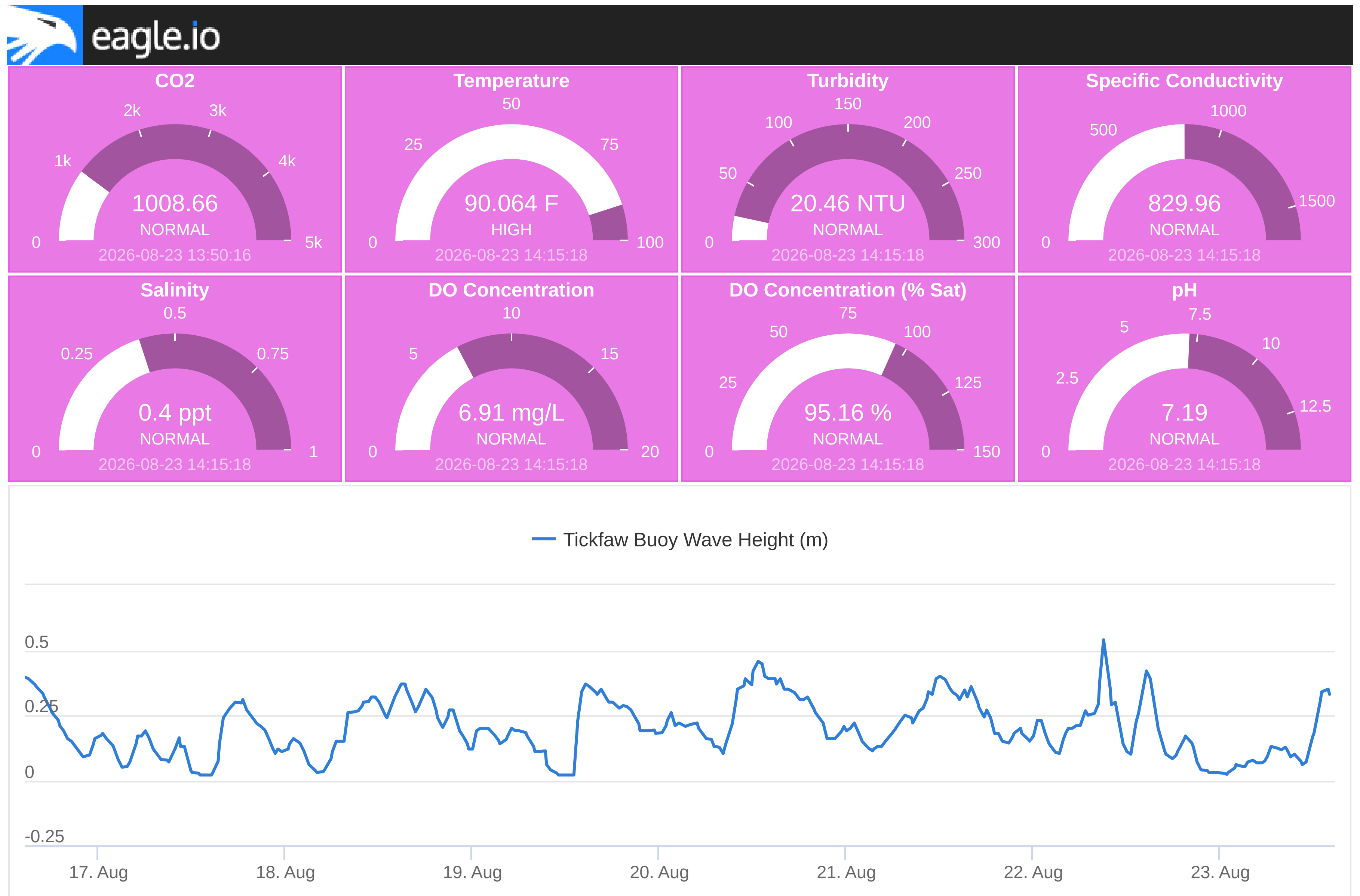
Historic wind speed and direction for the date range listed below. The frequency of winds over a time period is plotted by wind direction, with color bands showing wind speed ranges. Spoke length shows wind direction frequency.
[2026-07-23 14:43:50 - 2026-08-23 14:43:50]



Maurepas Buoy Water Quality Data



Tickfaw Buoy Water Quality Data



Water Chemistry Buoys

The water chemistry parameters monitored by these buoys include temperature, turbidity, tryptophan, CDOM, refined fuels, BOD, COD, total coliform, and methane (CH_4, % volume and mg/L). For more information regarding the parameters monitored by these buoys, see the table below. All water chemistry parameters are measured by all 7 buoys every 10 minutes, starting on the date of deployment (9/23/2025).

Temperature

Units: Celsius (C)

Description: Temperature of the water.

Turbidity

Units: Nephelometric Turbidity Unit (NTU)

Description: The opaqueness of the water due to the presence of suspended solids.

Tryptophan

Units: Parts per billion (ppb)

Description: Tryptophan fluorescence is used as an indicator of biological activity and organic matter inputs, including potential wastewater or nutrient enrichment. Temperature corrected.

CDOM

Units: Parts per billion (ppb)

Description: Chromophoric Dissolved Organic Matter (CDOM) influences water color, light penetration, and can indicate the presence of decaying plant material and other organic matter. Temperature corrected.

RefFuel

Units: Parts per billion (ppb)

Description: Refined fuels. An estimated concentration of refined petroleum hydrocarbons in water, expressed in parts per billion (ppb). Elevated values may indicate the presence of fuels or other refined oil products on the water surface.

BOD

Units: Milligrams per liter (mg/L)

Description: Biochemical Oxygen Demand (BOD). An algorithm-based estimate of the oxygen required by microorganisms to break down organic matter in water. Higher values generally indicate higher organic pollution and greater oxygen demand.

COD

Units: Milligrams per liter (mg/L)

Description: Chemical Oxygen Demand (COD). An algorithm-based estimate of the oxygen required to chemically oxidize organic matter in water. Higher values generally indicate higher levels of organic pollution.

Total Coli

Units: Colony-forming units per 100 milliliters (CFU/100 mL)

Description: Total Coliform (Total Coli). An algorithm-based estimate of total coliform bacteria concentrations in water derived from fluorescence measurements. Higher values generally indicate increased bacterial presence and potential impacts from wastewater, runoff, or other sources of organic contamination.

Methane

Units: Percent volume (%)

Description: Dissolved Methane (CH_4). The concentration of methane dissolved in water expressed as a percentage of volume. Elevated methane concentrations may indicate increased anaerobic microbial activity, sediment carbon processing, or external methane inputs.

Methane

Units: Milligrams per liter (mg/L)

Description: Dissolved Methane (CH_4). The concentration of methane dissolved in water expressed as milligrams per liter. Elevated methane concentrations may indicate increased anaerobic microbial activity, sediment carbon processing, or external methane inputs.